

each one was. Because the data is stored in the GeoJSON format, we'll work with it using the `json` module. Using Plotly's `scatter_geo()` plot, you'll create visualizations that clearly show the global distribution of earthquakes.

Downloading Earthquake Data

Make a folder called `eq_data` inside the folder where you're saving this chapter's programs. Copy the file `eq_1_day_m1.geojson` into this new folder. Earthquakes are categorized by their magnitude on the Richter scale. This file includes data for all earthquakes with a magnitude M1 or greater that took place in the last 24 hours (at the time of this writing). This data comes from one of the United States Geological Survey's earthquake data feeds, at <https://earthquake.usgs.gov/earthquakes/feed>.

Examining GeoJSON Data

When you open `eq_1_day_m1.geojson`, you'll see that it's very dense and hard to read:

```
{ "type": "FeatureCollection", "metadata": { "generated": 1649052296000, ...
  { "type": "Feature", "properties": { "mag": 1.6, "place": "63 km SE of Ped...
  { "type": "Feature", "properties": { "mag": 2.2, "place": "27 km SSE of Ca...
  { "type": "Feature", "properties": { "mag": 3.7, "place": "102 km SSE of S...
  { "type": "Feature", "properties": { "mag": 2.92000008, "place": "49 km SE...
  { "type": "Feature", "properties": { "mag": 1.4, "place": "44 km NE of Sus...
--snip--
```

This file is formatted more for machines than humans. But we can see that the file contains some dictionaries, as well as information that we're interested in, such as earthquake magnitudes and locations.

The `json` module provides a variety of tools for exploring and working with JSON data. Some of these tools will help us reformat the file so we can look at the raw data more easily before we work with it programmatically.

Let's start by loading the data and displaying it in a format that's easier to read. This is a long data file, so instead of printing it, we'll rewrite the data to a new file. Then we can open that file and scroll back and forth through the data more easily:

```
eq_explore    from pathlib import Path
_data.py      import json

# Read data as a string and convert to a Python object.
path = Path('eq_data/eq_data_1_day_m1.geojson')
contents = path.read_text()
❶ all_eq_data = json.loads(contents)

# Create a more readable version of the data file.
❷ path = Path('eq_data/readable_eq_data.geojson')
❸ readable_contents = json.dumps(all_eq_data, indent=4)
path.write_text(readable_contents)
```
